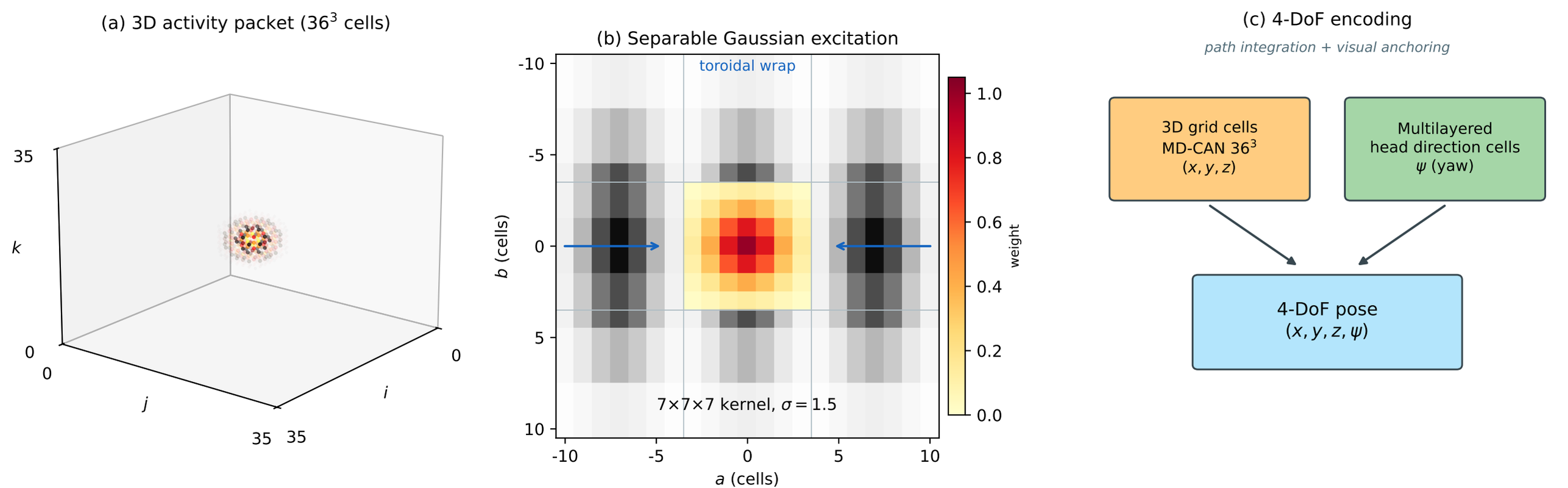

(a) 3D activity packet (36^3 cells)
35
k
0
0
j
35
35
i
0
(b) Separable Gaussian excitation
toroidal wrap
-10
-5
0
5
10
b (cells)
7×7×7 kernel, σ = 1.5
-10
-5
0
5
10
a (cells)
1.0
0.8
0.6
0.4
0.2
0.0
weight
(c) 4-DoF encoding
path integration + visual anchoring
3D grid cells
MD-CAN 36^3
(x, y, z)
Multilayered
head direction cells
ψ (yaw)
4-DoF pose
(x, y, z, ψ)